

Mohamed Moustafa Dawoud

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Google Scholar: Mohamed Moustafa Dawoud

Research Interests

- My research focuses on **security, privacy, and responsible AI**, with a **human-centered** emphasis on how people understand, experience, and govern intelligent systems.
- **Human-centered security and privacy**
- **AI governance and accountable deployment**
- **Human–AI interaction, trust, and misuse**
- **Empirical and data-driven auditing**

Education

2025–Present **Ph.D. in Computer Science & Engineering**, *University of California, Santa Cruz*

Advisor: Prof. Ramakrishnan (Ram) Sundara Raman.

2021–2024 **B.Sc. in Informatics and Computer Science**, *German International University*, Cairo, Egypt

Full merit scholarship (top-ranked national students); 18th nationwide in Thanawya Amma (High School) Mathematics (~600,000 students), 408/410 (99.51%).

Ongoing Projects & Studies

Lead Researcher **AI Governance, Privacy & Regulation Study**, *UC Santa Cruz & Stanford University*

A qualitative study examining how organizations govern AI responsibly under emerging privacy and regulatory demands.

- Lead the full research lifecycle: IRB design, recruitment, interview protocol, data collection, and analysis.
- Conduct semi-structured 75-minute expert interviews to map governance structures, challenges, and emerging best practices.
- Align organizational practice with regulatory frameworks (CCPA/CPRA, GDPR, AI governance guidelines).
- Coordinate cross-institutional synthesis with RANDLab (UCSC) and Stanford's Empirical Security Research Group.

Publications

2026 **The Rise of NSFW Generative AI Services on Mainstream Marketplaces: A Case Study of Fiverr**, *Symposium on Usable Security and Privacy (USEC 2026)*, co-located with NDSS

Mohamed Moustafa Dawoud, Alejandro Cuevas, and Ramakrishnan (Ram) Sundara Raman.

Acceptance rate: 22/69 papers.

2025 **Vendor communication themes in darknet Ransomware-as-a-Service (RaaS) advertisements**, *Computers in Human Behavior*

Taylor Fisher, Zacharias Pieri, C. Jordan Howell, Roberta O'Malley, Lauren Tremblay, and **Mohamed Moustafa Dawoud**.

- 2024 **{DVA}: Extracting Victims and Abuse Vectors from Android Accessibility Malware**, *33rd USENIX Security Symposium (USENIX Security 24)*
Haichuan Xu, Mingxuan Yao, Runze Zhang, **Mohamed Moustafa Dawoud**, Jeman Park, and Brendan Saltaformaggio.
- 2022 **Social engineering and technical security fusion**, *2022 International Telecommunications Conference (ITC-Egypt)*
Wassim Alexan, Eyad Mamdouh, Mohamed ElBeltagy, Ahmed Ashraf, **Mohamed Moustafa Dawoud**, and Hashem Al-Qurashi.
- 2022 **Image Encryption Through Rössler System, PRNG S-Box and Recamán's Sequence**, *2022 IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC)*
Mohamed ElBeltagy, Wassim Alexan, Abdelrahman Elkhamry, **Mohamed Moustafa**, and Hisham H. Hussein.
- 2021 **IoMT Security: SHA3-512, AES-256, RSA and LSB Steganography**, *2021 8th NAFOSTED Conference on Information and Computer Science (NICS)*
Wassim Alexan, Ahmed Ashraf, Eyad Mamdouh, Sarah Mohamed, and **Mohamed Moustafa Dawoud**.

Research & Academic Experience

- Jun 2025–Present **Graduate Student Researcher**, *University of California, Santa Cruz*, Santa Cruz, CA, On-site
- Conduct research on human-centered security, privacy, and AI governance under Prof. Ramakrishnan (Ram) Sundara Raman, focusing on how organizations deploy, regulate, and manage risks in emerging AI systems.
 - Lead qualitative and mixed-methods studies of AI governance practices and AI-enabled abuse ecosystems, including expert interviews with industry practitioners and large-scale measurements of AI abuse-as-a-service on gig marketplaces.
 - Develop structured LLM-based evaluation frameworks for analyzing organizational AI and privacy policies and identifying gaps between stated governance expectations and real-world practices.
 - Designed a large-scale auditing pipeline integrating browser-based tracking measurements with rubric-guided LLM privacy-policy analysis to evaluate sector-wide compliance with modern privacy regulations (CCPA).
- Jan 2025–Jun 2025 **Graduate Teaching Assistant**, *Dartmouth College*, Hanover, NH, On-site
- Served as Lead Graduate Teaching Assistant for COSC 55: Security & Privacy (Prof. Sami Saydjari), supporting a class of 28 students in an advanced undergraduate course on computer and network security.
 - Designed and implemented a new AI-integrated teaching model that embraced tools such as ChatGPT and GitHub Copilot to foster critical thinking, explanation, and reflection rather than rote code production.
 - Created a new series of labs emphasizing system design freedom, oral reasoning, and justification of design choices; guided weekly one-on-one and team check-ins to assess conceptual understanding.
 - Conducted all grading, oral evaluations, and rubric design; emphasized rigorous yet supportive assessment through structured oral defenses.
 - Mentored students on trade-offs in assurance, usability, and privacy engineering—culminating in innovative final projects (e.g., a privacy-preserving, role-based messaging system with blind signature tokens).
 - Collaborated closely with the course instructor to co-develop teaching materials, improve pedagogical alignment, and evaluate how AI tools reshape student learning in computer security.
- Sep 2024–Dec 2024 **Research Associate**, *Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)*, Abu Dhabi, United Arab Emirates, On-site
- Conducted research on combating deepfakes and promoting the responsible use of AI through a human-centered lens.
 - Investigated how users form mental models and trust judgments when interacting with AI-generated deepfakes across visual and textual media.
 - Developed prototype systems and experimental frameworks for detecting AI-generated content and explaining authenticity decisions to end users.
 - Explored strategies for improving human–AI collaboration in misinformation detection and enhancing transparency and interpretability in AI-driven systems.

- Oct 2023–Feb 2024 **Research Intern, Georgia Institute of Technology, Atlanta, GA, On-site**
- Systems/network security (Prof. Brendan D. Saltaformaggio); contributions to work that appeared later at USENIX Security.
 - Investigated the potential of large language models to assist static analysis of Android malware.
 - Designed and executed a large-scale experiment using a manually curated dataset of benign and malicious Android Malwares' static code collected from the wild.
 - Demonstrated strong LLM capabilities for reasoning about malicious behaviors and summarizing complex code patterns; results submitted to the DIMVA conference.
- Jan 2023–Aug 2023 **Undergraduate Research Assistant, University of South Florida, Remote**
- Worked with Prof. C. Jordan Howell on cybercrime and underground economies, with a focus on Ransomware-as-a-Service (RaaS) marketplaces.
 - Led the threat-intelligence component of the project, including data collection and analysis of darknet RaaS vendor advertisements.
 - Contributed to the analysis that uncovered core communication patterns (e.g., victimization, persuasion strategies) used to market pre-built ransomware offerings.
 - This work appeared later as a peer-reviewed publication: "*Vendor Communication Themes in Darknet Ransomware-as-a-Service (RaaS) Advertisements*," *Computers in Human Behavior*.
- Sep 2021–Jan 2023 **Information Security Intern, German University in Cairo, Cairo, Egypt**
- With Prof. Wassim Alexan on applied security (cryptography, IoT/IoMT).

Service & Outreach

- 2025 **AAAS CASE Workshop Delegate**, Washington, DC. One of four graduate students selected to represent Dartmouth College at the *AAAS Catalyzing Advocacy in Science and Engineering (CASE) Workshop*; engaged with policymakers on science, technology, and public policy.
- 2023 **Conference Volunteer**, IEEE Secure Development Conference (IEEE SecDev 2023), Atlanta, GA. Supported session logistics and attendee coordination.
- Ongoing **Community Engagement & Mentorship**. Mentor prospective CS PhD applicants on research focus, materials, and strategy; featured in the *AAAS CASE 2025 Student Spotlight: Journey from Zefta to Dartmouth*.